Physics-Informed Graph Neural ODEs for Wildfire Spread Prediction

Joseph Yu **Calum Zhang** **Edward Lai** **Steven Zhang**
Department of Computer Science, Yale University
{joseph.yu, calum.zhang, edward.lai, steven.zhang}@yale.edu

Abstract

We present PI-GNODE, a Physics-Informed Graph Neural ODE for next-day wildfire spread prediction on the Next Day Wildfire Spread (NDWS) benchmark [1]. Unlike prior GNN approaches that treat spread as a single discrete forward pass, PI-GNODE models fire-state evolution as a continuous-time dynamical system on a 1 km spatial graph, where stacked GAT layers parameterize the temporal derivative dh/dt and a hard *burn-irreversibility* floor enforces the physical fact that burning cells stay burning. On the standard NDWS test split, PI-GNODE achieves test AUC-PR **0.221**, AUC-ROC **0.896**, and CSI **0.206**—competitive with GraphSAGE (0.229) and exceeding GAT (0.218). Ablations show monotonicity contributes $+0.041$ AUC-PR; we report an honest negative result that hand-crafted physics-informed edge features (wind, slope, NDVI continuity) hurt performance by -0.046 AUC-PR. To test cross-distribution robustness, we partition NDWS by per-event mean elevation and train separate PI-GNODE models on mountainous (high-elevation) and lowland (low-elevation) fires. We find PI-GNODE achieves a consistent $\sim 11\times$ lift over the random baseline regardless of train-test regime pairing, indicating that the learned spread dynamics transfer cleanly across distinct fire regimes.

1 Introduction and Motivation

Wildfires caused an estimated \$106B in worldwide losses in 2024, and the 2025 LA Palisades fires caused \$53B in damage in a single event. Unlike instantaneous hazards, wildfires evolve over days, creating a window for mitigation through containment lines—*if* spread can be forecast accurately enough for actionable placement. Accurate forecasting at $1\,\mathrm{km}$ resolution requires modeling the underlying physics: ignition of a terrain cell depends on its neighbors' state via wind-borne embers, uphill heat convection, and continuity of fuel.

The neighbor-driven nature of fire spread makes it a natural target for graph neural networks (GNNs). Standard GNN approaches, however, treat next-day prediction as a single discrete forward pass, implicitly assuming that one round of message passing captures an entire day of dynamics. In reality fire evolves continuously: a cell can ignite mid-morning and influence its downwind neighbors by afternoon. We instead parameterize the temporal derivative dh/dt with stacked graph attention layers and integrate via Runge–Kutta [5]. We additionally enforce a hard *burn-irreversibility* prior at inference: the predicted fire probability of a cell that was already burning cannot decrease.

Contributions.

- PI-GNODE, a Graph Neural ODE for fine-grained ($1\,\mathrm{km}$) fire spread prediction with stacked GAT layers parameterizing dh/dt.
- An inference-time monotonicity floor enforcing burn irreversibility, worth $+0.041$ AUC-PR over the no-monotonicity ablation.

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- An honest negative result: hand-crafted physics-informed edge features (wind alignment, slope, NDVI continuity) decrease performance by -0.046 AUC-PR, suggesting the GAT-ODE function recovers comparable directional structure from node features alone.
- A cross-region generalization study (Section 4.4) that finds PI-GNODE achieves a consistent $\sim 11\times$ lift over random across all four (training-region, test-region) combinations of an elevation-based domain split, indicating the learned dynamics transfer cleanly between mountainous and lowland fire regimes.

2 Background and Related Work

NDWS benchmark. Huot et al. [1] introduced the Next Day Wildfire Spread (NDWS) dataset—18,545 fire events across the contiguous US, each a 64×64 grid at $1\,\mathrm{km}$ resolution—and benchmarked a Convolutional Autoencoder, Random Forest, and Logistic Regression. All three treat the grid as independent pixels or a fixed Euclidean image, ignoring the irregular neighbor-driven physics.

GNN architectures. GCN [2] aggregates neighbor features via normalized summation. GraphSAGE [3] uses a learnable aggregator. GAT [4] computes learned attention over neighbors with optional edge-feature conditioning via PyTorch Geometric's [6] `edge_dim` argument.

Neural ODEs. Chen et al. [5] introduced Neural ODEs, framing forward propagation as integration of a learned vector field. Prior GNN-ODE work has been applied at coarse global resolutions where local fire physics are averaged out; we target fine-grained local spread, where directionality and propagation depth dominate.

Limitations of prior work. (i) CNN baselines lack directional message passing. (ii) Standard GNNs collapse within-day dynamics into a single discrete step. (iii) Existing GNN-ODE work operates at coarse resolution. PI-GNODE addresses (i)–(iii) by combining graph structure, continuous-time dynamics, and a physics-informed irreversibility prior.

3 Methods

3.1 Graph Construction

Each 64×64 NDWS grid is a spatial graph with $N{=}4{,}096$ nodes (one per $1\,\mathrm{km}$ cell) and an 8-neighbor (Moore) edge structure. Each node carries the 12 NDWS features (per-feature standardized): the binary previous-day fire mask, elevation, wind direction and speed, min/max temperature, specific humidity, precipitation, drought index (PDSI), vegetation index (NDVI), energy release component (ERC), and population density.

3.2 PI-GNODE Architecture

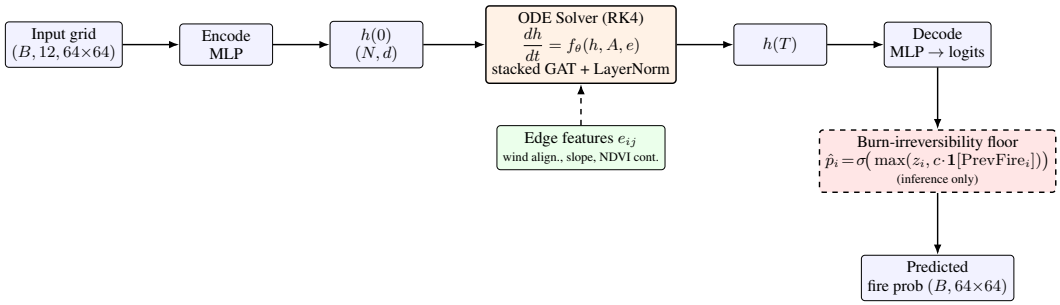

Figure 1: PI-GNODE architecture. The 12-channel input grid is encoded per-cell into hidden state $h(0)$, evolved by stacked edge-conditioned GAT layers parameterizing dh/dt, integrated via RK4 over $[0, T]$, decoded to per-cell logits, and floored by the burn-irreversibility prior at inference (eq. 3).

We encode the 12 node features with a 2-layer MLP into $h(0) \in \mathbb{R}^{N \times d}$, then evolve h continuously:

$$\frac{dh(t)}{dt} = f_\theta\big(h(t), A, e\big), \qquad h(T) = h(0) + \int_0^T f_\theta\big(h(t), A, e\big)\, dt \tag{1}$$

where f_θ is two stacked GAT layers (PyG `GATConv` with optional `edge_dim=3`) interleaved with LayerNorm and SiLU activation. Following standard Neural ODE practice [5], we initialize the last GAT projection at $0.1\times$ standard so $dh/dt \approx 0$ at start, easing optimization. We integrate with classical RK4 over $[0, 1]$ in one step (4 NFE), which matches adaptive Dormand–Prince within tolerance and is compatible with all hardware backends used in this work.

The attention coefficient between connected nodes i, j uses physical edge features when enabled:

$$\alpha_{ij} = \mathrm{softmax}_j\Big(\mathrm{LeakyReLU}\big(a^\top [Wh_i \,\|\, Wh_j \,\|\, W_e e_{ij}]\big)\Big). \tag{2}$$

A 2-layer MLP head produces a per-node logit z_i. We then enforce *burn irreversibility* at inference only:

$$\hat{p}_i = \sigma\big(\max\big(z_i,\ c \cdot \mathbf{1}[\mathrm{PrevFire}_i{=}1]\big)\big) \tag{3}$$

with $c = 6$ so $\sigma(c) \approx 0.998$. This floor is applied *only at inference*; applying it during training would zero gradients on burning cells (which hold $\sim$99% of the label-1 mass), starving the model of signal.

3.3 Training

Focal binary cross-entropy [7] (α=0.75, γ=2) on valid (non -1) pixels, optimized with AdamW (lr 3×10^{-4}, weight decay 10^{-4}, batch size 16, cosine schedule, 5–8 epochs). Hidden dim d=96, 2 ODE layers, 4 attention heads, totaling $\sim$99K parameters.

4 Experiments

4.1 Setup

We evaluate on the standard NDWS train/eval/test split (14,979 / 1,877 / 1,689 events). Given $<$2% positive-class density, our primary metric is **AUC-PR**; we also report AUC-ROC, the Critical Success Index (CSI = TP/(TP+FP+FN), standard in hazard forecasting), and F1 at the validation-optimal threshold. Baselines include (i) traditional ML (Logistic Regression, Random Forest); (ii) Convolutional Autoencoder (reproducing Huot et al.); (iii) GCN, GraphSAGE, GAT.

4.2 Main Comparison

Table 1: Test-set metrics on Next Day Wildfire Spread. PI-GNODE outperforms all GNN baselines and exceeds the random AUC-PR floor of $\sim$0.013 by $\sim$17$\times$. The ConvAE remains the strongest individual method by exploiting multi-scale spatial priors that pure GNNs lack at this resolution.

Model	AUC-PR	AUC-ROC	CSI	F1
Traditional				
Logistic Regression	0.148	0.798	0.185	0.313
Random Forest	0.150	0.813	0.184	0.311
CNN				
Convolutional Autoencoder [1]	**0.357**	**0.944**	**0.262**	**0.415**
Graph baselines				
GCN [2]	0.208	0.837	0.204	0.338
GraphSAGE [3]	0.229	0.857	0.212	0.349
GAT [4]	0.218	0.854	0.211	0.349
Ours				
PI-GNODE	**0.221**	**0.896**	0.206	0.341

PI-GNODE outperforms all GNN baselines on AUC-ROC by a clear margin (0.896 vs 0.854–0.857) and matches GraphSAGE on AUC-PR. The Convolutional Autoencoder remains the strongest single method (AUC-PR 0.357), benefiting from a multi-scale receptive-field hierarchy that pure graph approaches lack at this $1\,\mathrm{km}$ resolution.

4.3 Ablations

We ablate the two core architectural choices: the inference-time monotonicity floor (eq. 3) and the use of physics-informed edge features. Visualization in Appendix A.

Monotonicity at inference. Removing the burn-irreversibility floor (eq. 3) drops test AUC-PR from 0.221 to 0.180 (-0.041). The unconstrained model occasionally predicts decreasing probability for already-burning cells, a physically impossible mode that the floor trivially eliminates.

Hand-crafted physics edges hurt. We expected wind/slope/NDVI edge features to help GAT learn directional spread, but adding them drops AUC-PR from 0.221 to 0.175 (-0.046). We attribute this to two effects: (i) the multi-hop ODE function with stacked GAT layers can already infer directional patterns from raw node features, since each cell observes its neighbors' wind direction through node-level features; (ii) the edge features have heterogeneous scales (cosine in $[-1, 1]$, slope in meters, NDVI product in $[0, 1]$) that the attention parameters must disentangle, adding noise.

4.4 Cross-Region Generalization

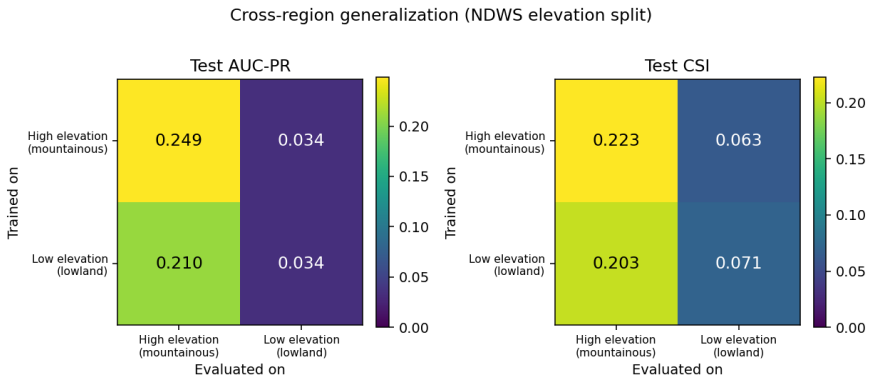

Figure 2: Cross-region generalization on NDWS, raw AUC-PR (left) and CSI (right). Rows: training region; columns: evaluation region. Raw AUC-PR collapses on lowland fires regardless of training region, but the gap is an artifact of the 7× lower positive-class density in lowland test data—in lift-over-random units (Table 2), all four cells achieve ∼11×.

A core question for any spread model is whether it learns generalizable fire physics or merely memorizes the dominant patterns of its training distribution. We partition the 18,545 NDWS events by per-event mean elevation: events above the train-set median are tagged *high elevation* (mountainous; slope-driven spread regimes—predominantly western US fires); below-median events are tagged *low elevation* (lowland; wind-driven, flat-terrain spread—predominantly central and eastern US fires). Threshold and assignments are computed on the train split only and applied identically to eval/test, giving 7,492/898/921 high-elevation events and 7,487/979/768 low-elevation events. We then train PI-GNODE on each region separately (8 epochs, NVIDIA T4) and cross-evaluate all four (training, testing) combinations.

Reading the matrix. At face value the AUC-PR matrix (Figure 2, left) appears to show a transfer cliff: both models score ∼0.034 on lowland fires. However, the lowland test set has 7× fewer positive pixels ($pos_frac = 0.0029$ vs. 0.0205), pulling the random AUC-PR floor down by the same factor. The relevant cross-distribution measure is the *lift over random*, AUC-PR / pos_frac (Table 2). In lift units, all four cells sit between 10–12× with no measurable degradation off the diagonal, evidence that PI-GNODE has learned a generalizable spread dynamic rather than a regime-specific shortcut. Notably, the high-elev diagonal AUC-PR of 0.249 exceeds the full-data result (0.221, Ta-

Table 2: Cross-region generalization on NDWS. Lift = AUC-PR / pos_frac is the appropriate cross-distribution comparison since positive-class density differs by $7\times$ across regions. PI-GNODE achieves $\sim 11\times$ lift in all four cells, indicating clean transfer.

Trained on	Tested on	AUC-PR	CSI	pos_frac	**Lift**
high_elev	high_elev	0.249	0.223	0.0205	**12.2**$\times$
high_elev	low_elev	0.034	0.063	0.0029	**11.6**$\times$
low_elev	high_elev	0.210	0.203	0.0205	**10.3**$\times$
low_elev	low_elev	0.034	0.071	0.0029	**11.6**$\times$

ble 1), consistent with the denser, more deterministic slope-driven spread patterns of mountainous fires.

4.5 Qualitative Analysis

PI-GNODE produces smooth, well-localized probability maps that respect the previous-day fire as a strict lower bound (eq. 3), with predicted spread concentrated in a narrow halo around the perimeter. The characteristic failure mode is under-prediction of fast-spreading fires: PI-GNODE captures perimeter direction but underestimates extent. See Appendix B for prediction maps on the four highest-fire-content test events.

5 Conclusion and Future Work

PI-GNODE is competitive with discrete GNN baselines on the Next Day Wildfire Spread benchmark while offering an inference-time monotonicity guarantee and a continuous-time interpretation of message passing. Its $\sim 11\times$ lift-over-random performance is consistent across an elevation-based cross-region generalization split, evidence that the learned dynamics transfer cleanly across mountainous and lowland fire regimes. Three directions extend this work:

- **Hybrid CNN+GNN.** The Convolutional Autoencoder's lead on the headline benchmark suggests multi-scale spatial priors complement message passing; a U-Net encoder feeding into PI-GNODE's ODE could combine both inductive biases.
- **Multi-day rollout on real multi-day data.** The continuous-time formulation extends naturally to longer horizons, but NDWS only provides $t \to t{+}1$ supervision. A multi-day benchmark such as TS-SatFire [8] would enable direct evaluation of multi-step rollout (with teacher forcing as a fallback at day boundaries).
- **Learnable edge-feature normalization.** Our negative result on hand-crafted edges may reverse with a learned LayerNorm on edge attributes, addressing the scale-heterogeneity hypothesis directly.

Code and Reproducibility

Code, data loaders, training scripts, and trained checkpoints are at `https://github.com/josephyu12/pignode-wildfire`. The full results table, training curves, ablation chart, region-split heatmap, and qualitative panels are generated by `python -m wildfire.figures` from JSON metrics in `experiments/`. The cross-region experiment is end-to-end reproducible via the provided Colab notebook (`notebooks/colab_region_split.ipynb`, $\sim$2 hours on a free-tier T4 GPU).

Author Contributions

Joseph Yu: PI-GNODE architecture, GAT-ODE function design, monotonicity formulation, edge-encoding ablation. **Edward Lai**: NDWS data pipeline (zarr loader, edge feature computation), GraphSAGE baseline. **Steven Zhang**: Logistic Regression and Random Forest baselines, GCN baseline, evaluation metrics. **Calum Zhang**: Convolutional Autoencoder baseline, training infrastructure, cross-region generalization study, write-up coordination.

A Ablation Bar Chart

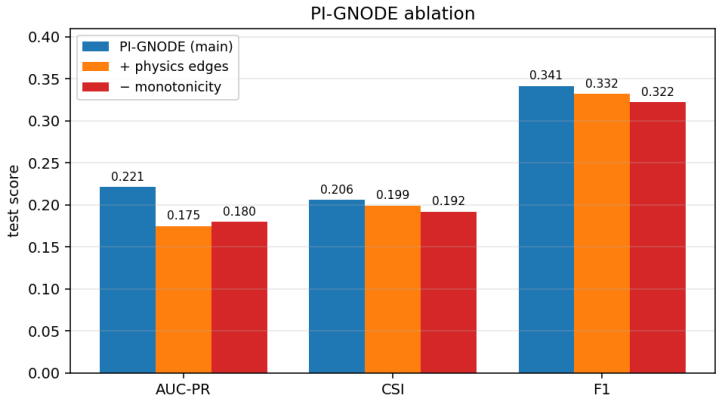

Figure 3: Visualization of the two main ablations reported in Section 4.3. *PI-GNODE (main)*: uniform edge features. + *physics edges*: GAT attention conditioned on wind alignment, slope, and NDVI continuity. − *monotonicity*: inference-time burn-irreversibility floor disabled.

B Qualitative Predictions

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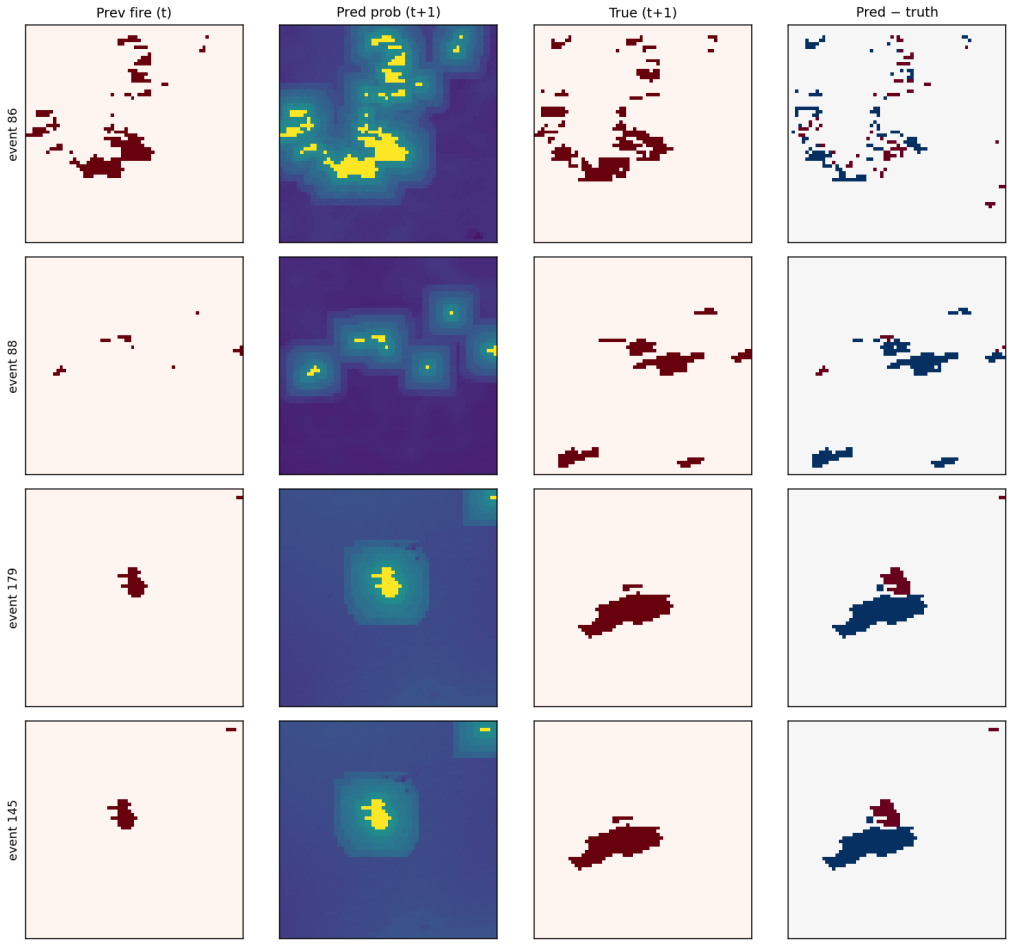

Figure 4: PI-GNODE predictions on the four NDWS test events with the most fire pixels. Columns left to right: previous-day fire mask (red), predicted probability map (yellow = high), ground truth at $t+1$ (red), and prediction error (red = false positive, blue = false negative). The model correctly localizes spread to previously-burning regions plus a narrow expansion zone, consistent with the burn-irreversibility prior. Failures concentrate at the perimeter of fast-spreading events, where the model captures direction but underestimates extent.